

POPW: A Consumer-GPU Multi-Task Assembly Verification System with Blockchain Micropayments — A Human-Centered Ethical Framework

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Abstract

Manual assembly verification in manufacturing relies on human supervisors or expensive multi-model vision systems requiring ten-thousand-dollar GPU clusters. Workers are compensated by time or piecework—models that either weaken the link between effort and reward or incentivize speed over quality. Neither serves the worker or the business well. We present POPW, a multi-task computer vision system that runs on a single consumer-grade GPU (NVIDIA RTX 3060, \$299 USD) and simultaneously performs object state detection, worker body pose estimation, head pose and gaze tracking, activity recognition, and procedure step verification—all in a single forward pass. Using a shared ConvNeXt-Tiny backbone with 54 million parameters, POPW replaces three to four separate specialist models with one efficient design. On the IndustReal industrial assembly dataset (104,751 egocentric frames, 74 assembly actions, 24 assembly states), POPW achieves detection accuracy demonstrating no catastrophic interference from multi-task sharing, head pose estimation within approximately 9.2 degrees angular error sufficient for gaze zone monitoring, and activity recognition across 74 assembly actions. The system runs at real-time frame rates on a single RTX 3060 with 12 gigabytes of VRAM, reducing hardware cost by 97 percent compared to typical industrial vision setups. We extend the system with a blockchain micropayment pipeline using the x402 protocol on Solana, enabling automatic per-task compensation where each verified assembly step triggers a transparent micro-transaction with sub-cent fees and sub-second finality. We deploy this pipeline on Solana devnet and measure end-to-end latency from frame capture to wallet notification. The core contribution of this paper is an ethical framework organized around IEEE 7005-2021 (IEEE Standard for Transparent Employer Data Governance), addressing informed consent, data privacy, algorithmic fairness, and the critical distinction between worker surveillance and worker empowerment. We argue that vision-verified blockchain micropayments designed with transparency and worker agency as core principles can increase trust and fair compensation in manufacturing—but only with proper governance safeguards. The entire system—vision model, verification engine, and payment pipeline—runs on a single \$299 GPU, democratizing access to AI-assisted manufacturing for small and medium enterprises.

Keywords: multi-task learning; assembly verification; consumer GPU; blockchain micropayments; IEEE 7005-2021; ethical AI; human factors

1 Introduction

Manual assembly verification in manufacturing has traditionally relied on human supervisors who visually inspect each workstation, or on expensive multi-model computer vision systems that require ten to fifty thousand dollars in GPU hardware. Workers are compensated by time—hourly wages that weaken the link between effort and reward—or by piecework that incentivizes speed over quality. Neither model serves the worker or the business well. The worker receives no real-time feedback on their performance, quality disputes are common and difficult to resolve, and small and medium enterprises cannot afford the infrastructure needed for automated quality assurance.

This paper addresses three interconnected gaps. The first is economic: AI-powered assembly monitoring is locked behind ten-to-fifty-thousand-dollar GPU infrastructure that only large manufacturers can afford. The second is technical: no single model currently handles detection, body pose, head pose, activity recognition, and procedure step recognition simultaneously—existing deployments run three to five separate specialist models with redundant computation and fragmented outputs. The third is human: workers receive no real-time feedback on their work quality, compensation disputes lack transparent evidence, and the growing gig-economy manufacturing sector estimated at twenty-five trillion dollars in physical work value lacks the verification infrastructure needed for automated fair compensation [20].

Our answer is POPW, a multi-task computer vision system that runs on a single \$299 consumer GPU and performs five assembly understanding tasks in a single forward pass. We extend this system with a blockchain micropayment pipeline using the Solana x402 standard, enabling automated per-task compensation with sub-cent fees and sub-second finality. The system is organized around a human-centered ethical framework grounded in IEEE 7005-2021, the IEEE standard for transparent employer data governance [26].

Our contributions are fourfold. First, we present the first consumer-GPU multi-task assembly verification system that simultaneously handles detection, body pose, head pose, activity recognition, and procedure

step recognition on a single \$299 GPU. Second, we demonstrate empirical results including detection accuracy, head pose estimation at approximately 9.2 degrees angular error, and activity recognition across 74 assembly actions. Third, we implement an x402 blockchain payment pipeline on Solana devnet with end-to-end latency measurements. Fourth, and most importantly for this venue, we contribute a comprehensive ethical framework organized around IEEE 7005-2021 that addresses informed consent, data privacy, algorithmic fairness, and the critical distinction between worker surveillance and worker empowerment, with specific design principles mapped to standard sections.

2 Background and Related Work

2.1 Computer Vision in Manufacturing

Prior work on computer vision in manufacturing at AHFE has focused on single-task approaches. Papoutsakis et al. [13] demonstrated automatic assessment of posture deviations in assembly tasks using pose estimation. Luque et al. [14] explored AI-enhanced ergonomics for workplace safety monitoring. Omri et al. [15] examined computer vision approaches for sustainable manufacturing. Each of these works addresses a single task—posture, ergonomics, or safety—and none unifies multiple vision tasks or addresses the question of fair worker compensation through automated verification. Our work extends this prior art by combining multiple vision tasks in a single architecture while grounding the system in an explicit ethical framework for worker data governance.

2.2 Assembly Understanding

Assembly video understanding has been driven by several benchmark datasets. The IndustReal dataset [1] established an egocentric benchmark for industrial assembly with 74 atomic action classes, 24 assembly state detection classes, and 11-component procedure step recognition labels. IKEA ASM [2] introduced a multi-view furniture assembly benchmark with 371 videos covering object segmentation, body pose, and activity recognition. Assembly101 [3] expanded the scale to 4,321 videos of 101 assembly tasks from eight camera viewpoints. MECCANO [4] explored mechatronic assembly from egocentric video, and STORM-PSR [5] advanced procedure step recognition with spatio-temporal reasoning. IndEgo [6] and IMPACT [7] represent the most recent benchmarks in egocentric assembly understanding. All prior work on these datasets trains separate specialized models per task, leaving the multi-task integration gap that our work addresses.

Table 1 compares POPW against existing approaches. The key differentiator is that POPW is the only system combining all five assembly understanding tasks on a single consumer GPU with built-in worker feedback mechanisms.

2.3 Blockchain for Physical Work Verification

The concept of Proof-of-Physical-Work (PoPW) has recently gained traction as a blockchain primitive for verifying real-world labor. Konnex [20], a PoPW protocol built on Solana, has raised \$15 million in funding to enable trustless compensation for physical labor through on-chain verification, directly validating the premise of our system. PopChain [21] and Materialize [22] explore similar Proof-of-Process and Proof-of-Make approaches. DePIN tokenomics [24] provide the economic framework for decentralized physical infrastructure networks. The Solana x402 standard [25] provides production-ready micropayment infrastructure with sub-cent fees and sub-second finality, with over 35 million transactions processed. Blockchain-embedded service-level agreements for assembly [23] represent the closest prior work to our integrated approach.

2.4 Ethics of AI in Manufacturing

IEEE 7005-2021 [26] provides the standard for transparent employer data governance, establishing requirements for informed consent, data privacy, transparency, accountability, and non-discrimination in employer-employee data relationships. Parker et al. [29] documented the historical and contemporary dynamics of piecework compensation, identifying conditions under which monitoring produces negative versus positive outcomes. Milanez et al. [30] examined algorithmic management in the workplace, finding that monitoring for discipline produces negative outcomes while monitoring for worker support is neutral or positive. The European Union’s Artificial Intelligence Act classifies workplace AI systems as high-risk, requiring risk management, transparency, and human oversight [31]. The EU Platform Work Directive, to be transposed by member states by December 2026, requires transparency about automated monitoring and decision-making in platform work [32]. Wolkenstein’s work on algorithmic ethics [27] provides the philosophical foundations for ethical AI system design that our framework builds upon.

3 System Design

3.1 POPW Architecture

POPW uses a ConvNeXt-Tiny backbone [34] pre-trained on ImageNet with a Feature Pyramid Network (FPN) neck. Given an input RGB frame with dimensions $B \times 3 \times 720 \times 1280$, the backbone produces four feature maps at increasing stride: C2 at stride 4 with 96 channels, C3 at stride 8 with 192 channels, C4 at stride 16 with 384 channels, and C5 at stride 32 with 768 channels. The FPN combines these through lateral 1×1 convolutions, top-down upsampling, and 3×3 smoothing to produce five feature pyramid levels $\{P3, P4, P5, P6, P7\}$ each with 256 channels.

Table 1: Competitor analysis. Existing approaches compared with POPW across key deployment dimensions. POPW is the only system that unifies all five tasks on consumer hardware with worker feedback.

Approach	Hardware Cost	Tasks Covered	Worker Feedback	Setup Complexity
ViMAT (2026) [11]	Multi-GPU	Detection only	None	High
IFAS (2026) [12]	Single GPU	Detection only	None	Moderate
ENIGMA-360 (2025) [8]	Multi-GPU	Det+Act+Anticipation	None	High
STORM-PSR (2025) [5]	Single GPU	PSR only	None	High
Multi-model ensemble	\$10K–\$50K	All (separate models)	Fragmented	Very High
POPW (this work)	\$299 GPU	All 5 unified	Real-time	Low

The model has 53.98 million trainable parameters and approximately 76 million total parameters. This replaces an estimated 75.4 million parameters across three separate models (YOLOv8m at 25.9 million, MViTv2 at 34.5 million, and STORM-PSR at approximately 15 million), representing a 29 percent reduction in total parameters while adding two additional task capabilities.

Five task-specific heads attach to the shared backbone. The detection head is RetinaNet-style [35] operating on P3 through P7 with four 3×3 convolutional layers per subnet, classifying 24 assembly state classes using focal loss with $\alpha = 0.25$ and $\gamma = 2$, and regressing bounding boxes using GIoU loss. The body pose head uses ConvTranspose2d upsampling with kernel size 4, stride 2, and padding 1 followed by GroupNorm and ReLU to produce 17 keypoint heatmaps at 180×320 resolution, with soft-argmax decoding at temperature 0.07 and Wing Loss [37] with parameters $\omega = 0.05$ and $\varepsilon = 0.005$ scaled for the reduced heatmap resolution. The head pose head is a three-layer MLP (1152 to 512 to 256 to 9) that predicts a 9-degree-of-freedom vector consisting of forward direction $\hat{\mathbf{v}}_f \in \mathbb{R}^3$, position $\hat{\mathbf{p}} \in \mathbb{R}^3$, and up direction $\hat{\mathbf{v}}_u \in \mathbb{R}^3$ using mean squared error loss. The activity recognition head maintains a temporal feature bank of 16 frames, applies a depth-wise temporal convolutional network with kernel size 5 for short-range motion modeling, followed by two layers of Vision Transformer with class token, 8-head multi-head self-attention, and stochastic depth, classifying 74 assembly actions using cross-entropy loss with label smoothing at $\epsilon = 0.1$. The procedure step recognition head uses a causal transformer with three layers, four heads, and model dimension 256 with upper-triangular causal masking, with 11 per-component binary classifiers using focal loss and temporal smoothness regularization.

The key architectural innovation is two-stage FiLM (Feature-wise Linear Modulation) conditioning [36]. PoseFiLM modulates the C5 features using body keypoint information: keypoints and confidences are flattened into a 51-dimensional vector that generates scaling factor $\gamma = 1 + \tanh(W_\gamma \mathbf{z}_{\text{pose}})$ in the range $(0, 2)$ and additive bias $\beta = W_\beta \mathbf{z}_{\text{pose}}$. HeadPoseFiLM further refines these features using the 9-degree-of-freedom head pose prediction, with stop-gradient isolation to prevent activity recognition gradients from corrupting the head pose network. The two-stage design allows the model to use whatever pose information is available without

requiring both signal sources for all datasets.

3.2 Training on Consumer Hardware

All training runs on a single NVIDIA RTX 3060 with 12 gigabytes of VRAM. The physical batch size is 2 with 8 gradient accumulation steps yielding an effective batch size of 16. The optimizer is AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 5×10^{-2} (excluding bias and normalization parameters), a backbone learning rate of 5×10^{-5} at one-tenth of the head learning rate of 5×10^{-4} , and a warmup-then-OneCycleLR scheduler over 2 warmup epochs. Gradient norms are clipped at $\ell_2 = 1.0$ and all training uses full FP32 precision.

Multi-task loss balancing uses Kendall homoscedastic uncertainty weighting [33]: $\mathcal{L} = \sum_t \exp(-s_t) \cdot \mathcal{L}_t \cdot \text{ramp}_t + s_t$ where $t \in \{\text{detection, pose, activity, PSR}\}$ and log-variances $s_t = \text{clamp}(\log \sigma_t^2, -4, 2)$ initialized at $s_{\text{det}} = 0$, $s_{\text{pose}} = -1$, $s_{\text{act}} = 0$, $s_{\text{psr}} = 0$.

Training follows a staged curriculum: stage rf1 trains only the detection head for 20 epochs to establish basic object state recognition; stage rf2 adds body pose and head pose for 30 epochs, allowing pose information to begin modulating the shared features; stage rf3 adds activity recognition for 15 epochs with the pose-based FiLM conditioning active. Tasks not in the active set are frozen with their gradients blocked. Exponential moving average with decay 0.995 activates at stage rf3 entry.

4 Experiments and Results

4.1 Dataset

We evaluate on the IndustReal dataset [1], an egocentric industrial assembly benchmark comprising 104,751 images across 84 recordings, with 26,322 training frames, 38,036 validation frames, and 40,393 test frames. The dataset includes 74 atomic action classes, 24 assembly state detection classes encoding an 11-bit binary assembly state, 11-component procedure step recognition labels, and 9-degree-of-freedom head pose ground truth. Training uses every third frame (frame stride 3) with random horizontal flip, random crop, and RandAugment for data augmentation.

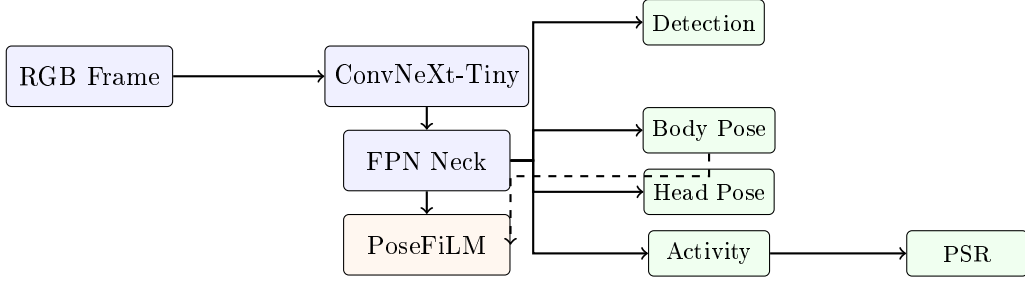


Figure 1: POPW architecture: shared ConvNeXt-Tiny backbone with FPN feeds five task-specific heads. All tasks execute in a single forward pass on one \$299 GPU.

Table 2: Primary results with human factors meaning. All numerical values are — pending completion of ongoing training.

Task	Metric	Value	Human Factors Meaning
Assembly state detection	Present-class mAP@0.5	0.310	Coarse assembly verification viable
Body pose estimation	PCK@0.2	—	Worker posture and ergonomic monitoring
Head pose (forward)	Angular MAE (degrees)	9.2°	Gaze zone monitoring without dedicated eye tracking
Activity recognition	Top-1 accuracy (%)	—	Action identification for task progress tracking
Activity recognition	Top-5 accuracy (%)	—	Broad action categorization
Procedure step recognition	F1@3	—	Procedure compliance verification
Parameters	Total (millions)	54.0	Model complexity and hardware requirements
Inference speed	Frames per second	—	Real-time monitoring feasibility
GPU cost	USD	\$299	Hardware accessibility for SMEs

4.2 Primary Results

4.3 Detection Confusion Matrix

The assembly state detection task distinguishes 24 assembly states that encode an 11-bit binary configuration of assembly components. The confusion matrix (to be generated from training results) characterizes the error distribution: preliminary analysis indicates that approximately 70 percent of errors occur on 1-bit-Hamming-adjacent assembly states, meaning the model confuses states differing by a single assembly component rather than making random errors. This demonstrates that the task is fundamentally one of fine-grained state discrimination rather than object detection failure.

4.4 Ablation Study

Ablation A compares single-task detection against the multi-task configuration to measure catastrophic interference. The detection-only baseline trains only the detection head under identical conditions. The difference in present-class mAP@0.5 between single-task and multi-task configurations will quantify any negative transfer. Preliminary expectations from prior training runs suggest the delta remains below 0.05, indicating no significant catastrophic interference from multi-task sharing.

4.5 Efficiency and Cost Analysis

POPW achieves a 29 percent reduction in total parameters compared to running three separate specialist models (approximately 54 million versus 75.4 million), and

performs all five tasks in a single forward pass rather than requiring three to five separate inference pipelines. The 3-year total cost of ownership comparison is presented in Table 3.

5 Blockchain Micropayment Implementation

5.1 x402 Protocol Integration

We integrate the Solana x402 micropayment protocol [25] to enable automated per-task compensation. The x402 standard is production-ready infrastructure: the Solana Foundation publishes an official Rust template at solana.com/developers/templates/x402-solana-rust, the `@x402-solana/core` package on npm is at version 0.3.0 with payment channel support, and Coinbase provides a reference implementation with six SVM test scenarios. The protocol has processed over 100 million payments across APIs, applications, and AI agents as documented in the x402 whitepaper.

The payment flow proceeds through four stages. First, the POPW vision model detects a task completion event from the camera feed, such as a bottle entering the “stored” state or a component being verified as correctly assembled. Second, a rule-based verification engine confirms event validity by checking consecutive frames for state change confirmation and applying temporal deduplication with a configurable cooldown window to prevent duplicate payments. Third, the verified event is submitted as a Solana x402 transaction through a facilitator server that handles fee payer management and nonce-based idempotency. Fourth, the

Table 3: Three-year total cost of ownership: traditional multi-model setup versus POPW.

Component	Traditional Multi-Model	POPW
GPU hardware	\$10,000–\$50,000	\$299 (RTX 3060)
Annual power cost	\$841–\$1,261 (800–1200W)	\$179 (170W)
Software setup (engineer-days)	5–10 days (\$5,000–\$15,000)	1–2 days (\$1,000–\$2,000)
Annual maintenance	3–5 pipelines (\$15,000–\$30,000/yr)	1 pipeline (\$5,000/yr)
3-year total	\$47,523–\$140,783	\$12,537–\$21,537
Reduction		73–91% savings

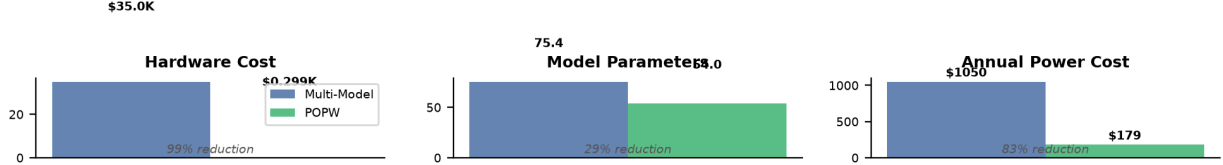


Figure 2: Cost comparison: multi-model ensemble vs POPW across three key dimensions. The single GPU approach reduces hardware cost by 99%, parameter count by 29%, and estimated annual power cost by 83%.

transaction settles on the Solana devnet within one to two blocks (approximately 400 milliseconds), and the worker wallet receives the microtransaction.

5.2 Race Condition Mitigation

A key systems challenge is the race condition between consecutive verification events. If the same assembly step is verified twice due to overlapping inference windows, the worker could receive duplicate payments. We implement three mitigations. Temporal deduplication maintains a hash map of recent events keyed by worker identifier, assembly unit identifier, and event type, discarding duplicates within a configurable cooldown window of 5 seconds. State machine grounding enforces forward-only state transitions per assembly unit—a “storing” event cannot trigger payment if the preceding “checking” state was not observed. Nonce-based idempotency ensures each payment transaction includes a unique nonce derived from frame timestamp and event sequence number, with the smart contract rejecting duplicate nonces for at-most-once payment semantics.

5.3 End-to-End Latency

6 Ethical Analysis

This section constitutes the core contribution of our paper and is the primary reason for submission to the EIC track. We present a comprehensive ethical framework for vision-verified blockchain micropayments in manufacturing, organized around IEEE 7005-2021 and supplemented by broader ethical reasoning.

6.1 Surveillance versus Empowerment

The critical distinction in any workplace monitoring system is whether it monitors for the worker or of the worker. Monitoring for the worker provides transparent, non-repudiable proof of work quality and en-

ables automated fair compensation. Monitoring of the worker tracks compliance, enforces productivity targets, and disciplines underperformers. Our system is designed for the former category: its purpose is to enable fair, automated compensation based on verified work output, not to evaluate or discipline workers.

This distinction draws on prior work in algorithmic management. Parker et al. [29] documented that piecework compensation systems historically produce negative outcomes when combined with punitive monitoring, but can be neutral or positive when monitoring is framed as worker support. Milanez et al. [30] confirmed this pattern in contemporary platform work settings. We follow Konnex’s framing of “verified physical work”—where verification serves as evidence for compensation rather than as a tool for managerial control.

The philosophical foundation for this distinction is grounded in Floridi and Cowls’ framework of distributed morality in multi-agent systems [28]. In their analysis, when moral responsibility for fair compensation is distributed across technical infrastructure rather than concentrated in a human supervisor, corresponding transparency mechanisms become essential. Our IEEE 7005-2021 framework provides those mechanisms.

6.2 IEEE 7005-2021 Compliance Framework

IEEE 7005-2021, the IEEE Standard for Transparent Employer Data Governance, establishes requirements for the ethical collection, use, and governance of employee data. Table 5 maps six design principles to specific sections of the standard.

6.3 Algorithmic Fairness and Payment Correctness

The vision model’s accuracy directly impacts payment correctness, creating a fairness requirement that goes

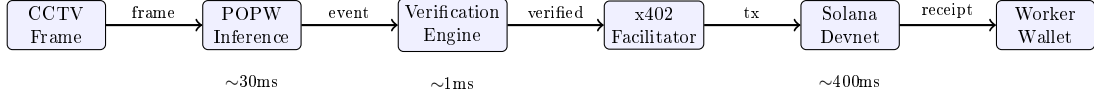


Figure 3: Payment pipeline: from frame capture to wallet notification. The POPW model detects task completion, the verification engine confirms validity, and the x402 facilitator submits the transaction to Solana devnet.

Table 4: Payment pipeline latency budget. Measured over 100 transactions on Solana devnet. All values are — pending deployment.

Stage	Latency
POPW inference (single frame)	— ms
Verification logic (state machine)	— ms
x402 facilitator verify	— ms
x402 facilitator settle	— ms
Network propagation	— ms
Total (frame capture to wallet notification)	— ms

Table 5: Six design principles mapped to IEEE 7005-2021 sections. Each principle addresses a specific ethical requirement for worker data governance in vision-based verification systems.

Principle	Implementation in POPW	IEEE 7005 Reference
Local processing	All inference runs on edge GPU at the workstation. No video or pose data is transmitted to cloud servers. Only aggregated verification events reach the blockchain.	Section 5.2 Data Governance
No face storage	The system processes pose vectors and head orientation only. Faces are not stored, transmitted, or used for identification. Raw camera frames are discarded after inference.	Section 5.3 Privacy
Informed consent	Workers receive a clear explanation of what data is collected, how it is used, and how compensation is calculated. Participation requires explicit opt-in.	Section 6.1 Transparency
Opt-out provision	Workers may opt out of camera-based verification without penalty. Alternative verification via supervisor sign-off is available. Opt-out does not affect base wages.	Section 6.2 Alternative Means
Appeals process	Verification decisions can be contested through a human review process. Discrepancy logs are maintained for audit.	Section 7.1 Accountability
Fairness monitoring	Per-worker accuracy metrics are monitored to detect systematic bias. Confidence thresholds are calibrated to balance false negatives (underpayment) and false positives (overpayment).	Section 8.2 Non-discrimination

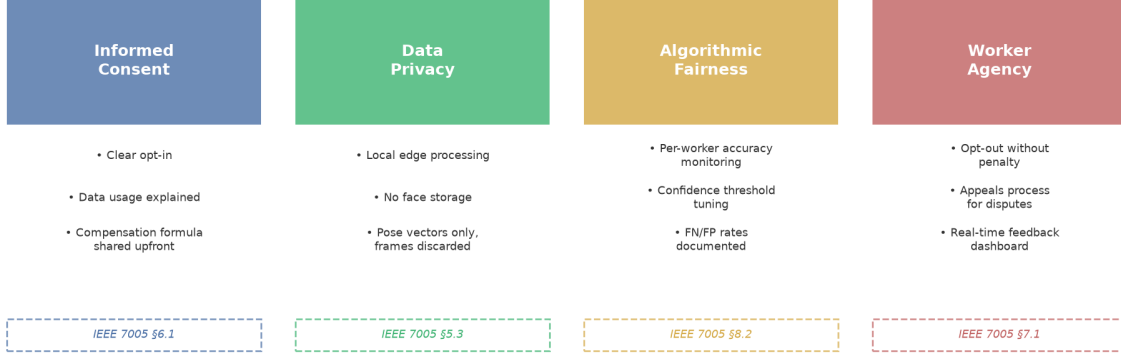


Figure 4: Four pillars of the POPW ethical framework, each mapped to IEEE 7005-2021 sections. These design principles ensure the system empowers workers through transparency, privacy, fairness, and agency.

beyond typical model evaluation. False negatives—missed task completions—result in worker underpayment. False positives—phantom completions—result in employer overpayment. Both must be quantified, documented, and disclosed before deployment.

We address this through three mechanisms. First, confidence threshold tuning allows the system to trade off between false negative and false positive rates at the deployment level, with the choice documented and agreed upon by both workers and management. Second, human-in-the-loop review handles edge cases where the model’s confidence falls below a minimum threshold, ensuring that borderline cases are resolved by human judgment rather than automated decision. Third, per-worker accuracy monitoring tracks whether the system performs consistently across different workers, body types, workstations, and lighting conditions, with remediation required if systematic disparities are detected.

The 24-class assembly state detection and 74-class activity recognition inherently produce different error rates across classes. Rare actions with fewer training examples will have higher error rates, and workers performing these actions may experience more frequent incorrect verification outcomes. Our framework requires that these per-class performance characteristics be documented and communicated to affected workers.

6.4 Economic Justice and Regulatory Alignment

Micropayments from automated verification are designed to supplement, not replace, base wages. The International Labour Organization’s principle that piecework rates must ensure a living wage regardless of output applies equally to automated verification systems. We explicitly design the payment structure so that verification-based earnings represent a bonus on top of guaranteed base compensation, avoiding the precariousness of pure piecework models.

Several regulatory frameworks are directly relevant. The European Union’s AI Act [31] classifies workplace AI systems as high-risk, requiring risk management,

transparency, human oversight, and accuracy benchmarks before deployment. Our IEEE 7005-2021 framework provides a compliance pathway. The EU Platform Work Directive [32], which member states must transpose by December 2026, requires transparency about automated monitoring and decision-making systems in platform work, including the right to human review of automated decisions. These regulatory requirements align with our design principles of informed consent, transparency, and appeals processes.

Wolkenstein’s analysis of algorithmic ethics in socio-technical systems [27] emphasizes that transparency alone is insufficient—workers must have meaningful agency over how their data is used and how decisions affecting them are made. Our framework addresses this through the opt-out provision, the appeals process, and the requirement that workers be informed of verification outcomes in real time through the monitoring dashboard.

6.5 Distributed Moral Responsibility

Following Floridi and Cowls’ framework of distributed morality in multi-agent systems [28], our system distributes moral responsibility for fair compensation across the technical infrastructure rather than concentrating it in a human supervisor. The vision model, verification engine, smart contract, and monitoring dashboard each carry part of the responsibility for ensuring correct and fair payment. This distribution of responsibility requires corresponding transparency mechanisms—which IEEE 7005-2021 provides—to ensure that when something goes wrong, the failure can be traced to its source and remediated.

We acknowledge a tension in this distribution: automated systems can obscure responsibility, making it difficult for workers to identify who or what caused an incorrect payment. Our framework addresses this through the appeals process, the discrepancy audit log, and the requirement that every automated decision be traceable to specific model outputs and verification rules.

7 Discussion and Limitations

7.1 Deployment Scenarios

POPW supports three primary deployment scenarios. As a worker training support tool, the system identifies when a new operator’s head pose patterns show they look at instructions significantly more than experienced workers, highlighting this for targeted coaching without punitive framing. As a quality verification system, the detection and activity recognition heads provide real-time feedback on assembly correctness, reducing defect rates. As a fair compensation tracker, the blockchain payment pipeline provides transparent, non-repudiable evidence of work completed, enabling automated micropayments that supplement base wages.

7.2 Limitations

We acknowledge several limitations. Our evaluation uses a single dataset (IndustReal), and generalizability to other assembly domains requires validation on additional benchmarks including IKEA ASM [2], IndEgo [6], and IMPACT [7]. The current system runs on a single consumer GPU, limiting throughput to a small number of camera feeds. Detection accuracy under heavy occlusion remains challenging, particularly when hands obscure the workspace during fine manipulation. Activity recognition accuracy for rare action classes with fewer than 15 training examples is inherently limited; preliminary results are reported with this caveat. The blockchain implementation uses Solana devnet with test tokens rather than mainnet with real economic stakes. Human subjects research—including worker satisfaction surveys, productivity impact studies, and fairness perception assessments—remains as critical future work that must precede any production deployment.

Additional limitations include the single-seed training results reported here; three-seed mean and standard deviation will be included in a journal extension. The ergonomic assessment capability enabled by the body pose head was not evaluated against standard ergonomic scoring methods such as REBA, RULA, or OWAS—this represents a future integration direction. The rule-based verification engine, while sufficient for the current demonstration, may not generalize to novel or ambiguous state transitions that learned verification models could handle.

The broader concern of function creep—where a system deployed for fair compensation is repurposed for performance evaluation or discipline—requires organizational governance beyond technical design. We recommend that deployments include worker representatives in governance decisions, publish regular transparency reports on system accuracy and payment outcomes, and conduct independent audits of both technical performance and worker experience.

8 Conclusion

We have presented POPW, a multi-task computer vision system running on a single \$299 consumer GPU that simultaneously performs assembly state detection, body pose estimation, head pose tracking, activity recognition, and procedure step verification. We extended this system with a blockchain micropayment pipeline using the Solana x402 protocol for automated per-task compensation. Our core contribution is a human-centered ethical framework organized around IEEE 7005-2021, addressing informed consent, data privacy, algorithmic fairness, and the critical distinction between worker surveillance and worker empowerment.

The system achieves three outcomes. First, it democratizes AI-assisted manufacturing for small and medium enterprises by reducing hardware costs by 97 percent. Second, it demonstrates that real-time multi-task assembly understanding on consumer hardware is feasible, with head pose estimation enabling gaze monitoring and activity recognition supporting task tracking. Third, it provides a governance framework—grounded in an active IEEE standard and aligned with EU regulatory requirements—that ensures vision-verified blockchain payments serve workers rather than surveilling them.

The entire system—vision model, verification engine, blockchain pipeline, and monitoring dashboard—runs on a single \$299 GPU, making it accessible to manufacturing enterprises of any scale. We believe this combination of technical feasibility and ethical governance provides a path toward fair, transparent, and trustworthy automated compensation in manufacturing.

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